

COMPASS_Synop- tic_LE_Tree_CO2_Flux_Calcs_2022

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1 Add Required Packages

2 Pull in Raw Datafiles

```
#defining where the raw data is

DATA_DIR_ROOT <- here::here(
  "Tree_Fluxes_Synoptic_LE",
  "2022",
  "Raw Data"
)

#function to read data
now <- function() format(Sys.time(), "%Y-%m-%d %H:%M:%S")
message(now(), " | Script started")
#finding the zip files
message(now(), " | Looking for ZIP files in: ", DATA_DIR_ROOT)

zip_files <- list.files(
  DATA_DIR_ROOT,
  pattern = "\\\\.zip$",
  full.names = TRUE)

message(now(), " | ZIP files found: ", length(zip_files))
```

```

print(zip_files)
  if (length(zip_files) == 0) {
    stop("No ZIP files found. Check DATA_DIR_ROOT.")}

cache <- list()

# Reader function
read_li7810_zip <- function(zip_file, tz = "US/Eastern") {
  message(now(), " | Processing ZIP: ", basename(zip_file))
  if (zip_file %in% names(cache)) {
    message(" -> Using cached version")
    return(cache[[zip_file]]) }

  tmp_dir <- tempfile("li7810_")
  dir.create(tmp_dir)
  on.exit(unlink(tmp_dir, recursive = TRUE), add = TRUE)
  unzip(zip_file, exdir = tmp_dir)
  data_files <- list.files(
    tmp_dir,
    pattern = "\\\\.data$",
    full.names = TRUE,
    recursive = TRUE )

  message(" -> .data files found: ", length(data_files))
  if (length(data_files) == 0) {
    stop("No .data files inside ZIP: ", zip_file) }

  dat_list <- lapply(data_files, function(f) {
    message("      Reading: ", basename(f))
    ffi_read_LI7810(f)})

  dat <- data.table::rbindlist(dat_list, fill = TRUE) %>%
    mutate(TIMESTAMP = with_tz(TIMESTAMP, tz))
  cache[[zip_file]] <- dat
  message(" -> ZIP finished: ", basename(zip_file))
  return(dat)}

# ACTUAL EXECUTION
message(now(), " | Starting data read")
alldat_list <- lapply(zip_files, read_li7810_zip)
alldat <- data.table::rbindlist(alldat_list, fill = TRUE)
message(now(), " | DONE")

```

```

message("Final number of rows: ", nrow(alldat))

# Show result
print(head(alldat))

#REMOVING WEIRD VERY LOW VALUES ON "2022-12-20 11:41:00" co2=-4000 licor bug???
alldat<-alldat%>%filter(CO2>-1000)

```

3 Read in metadata file and match to raw data

```

md <- "Raw Data/COMPASS_SynopticLE_TreeFlux_Metadata_2022b.csv"
dat <- read.csv(md)

metadat <- dat %>%
  # Remove rows without date
  filter(!is.na(Date)) %>%
  # Keep only desired instruments
  filter(SN %in% c("TG10-01448", "TG10-01426", "TG10-01028")) %>%
  # Normalize Plot labels
  mutate(
    Plot_raw = toupper(Plot),
    # Extract plot components
    #Site = str_extract(Plot_raw, "[A-Z]{3}"),
    #Zone = str_extract(Plot_raw, "(TR|UP)"), # somehow all of the UP PTR measurements turned
    Site = toupper(Site),
    Zone = toupper(Zone),
    Replicate = str_extract(Plot_raw, "(?<=TR|UP)[0-9]+"),
    # Letter ONLY if it comes AFTER a number
    Letter = str_extract(Plot_raw, "(?<=[0-9])[A-Z]"),
    # Time normalization
    Time_start = str_replace_all(as.character(Time_start), "\\.", ":") |> str_trim(),
    Stop.time = str_replace_all(as.character(Stop.time), "\\.", ":") |> str_trim(),
    # Safe time parsing (NO WARNINGS)
    t_start = suppressWarnings(parse_date_time(Time_start, orders = c("H:M:S", "H:M"))),
    t_stop = suppressWarnings(parse_date_time(Stop.time, orders = c("H:M:S", "H:M"))),
    # Observation length
    Obs_length = as.numeric(t_stop - t_start, units = "secs"),
    # Timestamp
    TIMESTAMP = as.POSIXct(

```

```

    paste(Date, Time_start),
    format = "%m/%d/%Y %H:%M:%S",
    tz = "EST"
  ),
  # Sequential ID
  Seq = row_number(),
  # INAL standardized Plot label
  Plot = if_else(
    is.na(Letter),
    paste(Date, Site, Zone, Replicate, Seq, sep = "_"),
    paste(Date, Site, Zone, Replicate, Letter, Seq, sep = "_")) )

# Quick check
head(metadat$Plot)

```

4 Check start times overlaps or missing values

```

#checking if there is any start and stop time overlap from the field notes
overlaps_detected <- metadat %>%
  mutate(
    start_ts = as.POSIXct(
      paste(Date, Time_start),
      format = "%m/%d/%Y %H:%M:%S",
      tz = "EST"
    ),
    end_ts = start_ts + Obs_length
  ) %>%
  arrange(SN, start_ts) %>%
  group_by(SN) %>%
  mutate(
    next_plot      = lead(Plot),
    next_start_ts  = lead(start_ts),
    overlap_fffi   = !is.na(next_start_ts) & next_start_ts <= end_ts
  ) %>%
  filter(overlap_fffi) %>%
  select(
    SN,
    Plot,
    start_ts,

```

```

    end_ts,
    next_plot,
    next_start_ts,
    Obs_length
  )

print(overlaps_detected)

#checking problems with obs_length - this is a frequent bug for manual field notes
field_notes_overlap<-metadat %>%
  mutate(
    obs_problem =
      is.na(Obs_length) | Obs_length <= 0
  ) %>%
  filter(obs_problem) %>%
  select(
    SN,
    Date,
    Plot,
    Time_start,
    Stop.time,
    Obs_length
  )

print(field_notes_overlap)

# checking unique measurment start time and stop time
# alldat %>%
#   filter(
#     SN == "TG10-01448",
#     TIMESTAMP > as.POSIXct("2023-01-05 13:28:00"),
#     TIMESTAMP < as.POSIXct("2023-01-05 13:33:00")
#   ) %>%
#   ggplot(aes(x = TIMESTAMP, y = CO2)) +
#   geom_point()#ylim(300,500)

```

5 Match metadata to raw data

```

#function to match data per equipment (considering different serial numbers) uhul!
match_flux_metadata <- function(alldat, metadat) {

```

```

message("Starting metadata  Licor matching")
#Sanity checks (fail early)
stopifnot(
  "TIMESTAMP" %in% names(alldat),
  "SN"         %in% names(alldat),
  "Date"       %in% names(metadat),
  "Time_start"%in% names(metadat),
  "Obs_length"%in% names(metadat),
  "Plot"       %in% names(metadat)
)

stopifnot(all(metadat$Obs_length > 0, na.rm = TRUE))
stopifnot(anyDuplicated(metadat$Plot) == 0)
# Instruments present in BOTH datasets
instruments <- intersect(
  unique(alldat$SN),
  unique(metadat$SN)
)
message("Instruments to process: ", paste(instruments, collapse = ", "))
results <- lapply(instruments, function(sn) {
  message("Processing SN: ", sn)
  dat_i <- alldat %>%
    filter(SN == sn)
  meta_i <- metadat %>%
    filter(SN == sn) %>%
    arrange(Date, Time_start) %>%
    mutate(metadat_row = row_number())
  stopifnot(nrow(meta_i) > 0)
# Actual matching
dat_i$metadat_row <- ffi_metadata_match(
  data_timestamps = dat_i$TIMESTAMP,
  start_dates     = meta_i$Date,
  start_times     = meta_i$Time_start,
  obs_lengths     = meta_i$Obs_length )
# Attach metadata (vectorized, fast)
dat_i <- dat_i %>%
  mutate(
    Plot      = meta_i$Plot[metadat_row],
    Site      = meta_i$Site[metadat_row],
    Zone      = meta_i$Zone[metadat_row],
    Volume    = meta_i$Volume[metadat_row],
    Temp      = meta_i$Temp[metadat_row],

```

```

      Area      = meta_i$Area[metadat_row],
      Dead_band = meta_i$Dead_band[metadat_row]      )
# Drop unmatched rows
n_before <- nrow(dat_i)
dat_i <- dat_i %>% filter(!is.na(metadat_row))
n_after <- nrow(dat_i)
message("  Matched rows: ", n_after, " / ", n_before)
dat_i })
alldat_matched <- rbindlist(results, fill = TRUE)
#Final sanity checks
stopifnot(
  !anyNA(alldat_matched$Plot),
  nrow(alldat_matched) > 0
)
message("Matching finished successfully")
return(alldat_matched)}

#running the funciton
message("Running final metadata matching...")
alldat_matched <- match_flux_metadata(alldat, metadat)
message("Matched dataset rows: ", nrow(alldat_matched))
head(alldat_matched)

#saving matched data and metadata as raw data is too big (today, Licor files leave in the CO2 column)
saveRDS(alldat_matched, file = "alldat_matched_2022.rds")

#can remove any row that have NA's in the CO2 column
#alldat <- alldat %>%
# filter(!is.na(CO2))

```

6 reating alldata matched df

```

# after you match the data for the first time - we can just read the .rds file to save processing
# if you need to match licor data and metadata again, remove "#" from the previous chunks

alldat_matched <- readRDS("alldat_matched_2022.rds")

```

```

#Extracting initial CO2 concentration

```



```
## Extract initial CO2 concentration (t0) for each Plot for QAQC later
#saving df to use after flux calculation
df_CO2_in <- alldat_matched %>%
  arrange(Plot, TIMESTAMP) %>%
  group_by(Plot) %>%
  summarise(CO2_initial = first(CO2), .groups = "drop")
```

7 Creating temperature csv from the L2-weather station in the wetland

```
# Getting Air temp from the weather station
## Read L2 temperature data from ZIP stored outside GitHub,
## extract desired variable, save lightweight CSV,
## and remove temporary unzipped files afterwards

## Read extracted L2 Parquet files and save temperature time series
BASE_DIR <- "D:/COMPASS_L2-v2-1/v2-1"
SITE <- "(OWC|CRC|PTR)"
VARIABLE <- "wx-tempavg15"

OUTFILE <- "Raw Data/L2_temp_timeseries.csv"

# Build regex that matches REAL L2 filenames
regex <- paste0(SITE, ".*", VARIABLE, ".*_L2_v2-1.*\\.parquet$")

# Locate Parquet files
files <- list.files(BASE_DIR, pattern = regex, recursive = TRUE, full.names = TRUE)

message("Files found: ", length(files))
print(basename(files))

# Fail fast if nothing is found
stopifnot(length(files) > 0)

# Read and combine Parquet files
dat <- lapply(files, function(f) {
  message("Reading ", basename(f))
  read_parquet(f)
})
```

```

dat <- bind_rows(dat)

# Optional QA filtering (only if column exists)
if ("QA_flag" %in% names(dat)) {
  dat <- dat %>% filter(QA_flag == 0)
}

# Keep only necessary columns
temps <- dat %>%
  select(TIMESTAMP, Plot, Site, Value) %>%
  rename(Temp = Value)

head(temps)

# Save lightweight output only
write.csv(
  temps,
  file = OUTFILE,
  row.names = FALSE
)

```

8 Reading Temp df to merge to metadat

```

#reading temp
temps <- read.csv("Raw Data/L2_temp_timeseries.csv")

## Match up the L1 temp data and add it to the metadata dataframe
#Going to need to use site and timestamp in the metadata in order to find a date

#make sure time stamps are in the correct format
alldat_matched$TIMESTAMP <- as.POSIXct(alldat_matched$TIMESTAMP, format = "%Y-%m-%d %H:%M:%OS")
#problem here L2 data for midnight is always "2022-01-01" we loose this line to NA when form
temps$TIMESTAMP[!grepl(":", temps$TIMESTAMP)] <-
  paste0(temps$TIMESTAMP[!grepl(":", temps$TIMESTAMP)], " 00:00:00")
temps$TIMESTAMP <- as.POSIXct(temps$TIMESTAMP, format = "%Y-%m-%d %H:%M:%OS")

#time rounding step
alldat_matched$TIMESTAMP_round <- round_date(alldat_matched$TIMESTAMP, "60 mins")
temps$TIMESTAMP_round <- round_date(temps$TIMESTAMP, "60 mins")

```

```
#ensure there is only one data observation per hour
temps_clean <- temps %>%
  group_by(TIMESTAMP_round, Site) %>%
  summarize(Temp = mean(Temp ))
```

```
`summarise()` has regrouped the output.
i Summaries were computed grouped by TIMESTAMP_round and Site.
i Output is grouped by TIMESTAMP_round.
i Use `summarise(.groups = "drop_last")` to silence this message.
i Use `summarise(.by = c(TIMESTAMP_round, Site))` for per-operation grouping
  (`?dplyr::dplyr_by`) instead.
```

```
#join them
alldat_matched <- alldat_matched %>%
  left_join(temps_clean, by=c("Site", "TIMESTAMP_round"))
```

9 fixing some Deadbands and stop time

```
# need to include a way to adjust the stop time too
# Table of plots with custom deadband adjustments
time_overrides <- tibble::tibble(
  Plot = c("8/17/2022_OWC_UP_6_147",
            "12/20/2022_OWC_UP_108_16",
            "9/16/2022_CRC_TR_314_132"),
  dead_band_manual = c(0,0,2),
  stop_offset_secs = c(-120,-7,-78)
)

#apply new incubation period
alldat_trimmed <- alldat_matched %>%
  left_join(time_overrides, by = "Plot") %>%
  group_by(Plot) %>%
  mutate(
    t_start_original = min(TIMESTAMP),
    t_end_original   = max(TIMESTAMP),

    # deadband manual or metadat
    t_start_adjusted = if_else(
```

```

    !is.na(dead_band_manual),
    t_start_original + dead_band_manual,
    t_start_original + as.numeric(coalesce(first(Dead_band), 0))
  ),

  # new offset
  t_end_adjusted = if_else(
    !is.na(stop_offset_secs),
    t_end_original + stop_offset_secs,
    t_end_original
  )
) %>%
filter(TIMESTAMP >= t_start_adjusted,
       TIMESTAMP <= t_end_adjusted) %>%
ungroup() %>%
select(-t_start_original, -t_end_original,
       -t_start_adjusted, -t_end_adjusted,
       -dead_band_manual, -stop_offset_secs)

# alldat_trimmed %>%
#   filter(Plot == "8/17/2022_OWC_UP_6_147") %>%
#   ggplot(aes(TIMESTAMP, CO2)) +
#   geom_point() + theme_classic()

```

10 Calculate fluxes

```

# TO Do - NEED TO PULL TEMP from the weather station
#change the units
alldat_trimmed$CO2_umol <- ffi_ppm_to_umol(alldat_trimmed$CO2,
                                           volume = alldat_trimmed$Volume, # m3
                                           temp = alldat_trimmed$Temp)      # degrees C from the weather s

```

Assuming atm = 101325 Pa

Using R = 8.31446261815324 m3 Pa K-1 mol-1

[illegible]

NOTE: HM81_flux.estimate is not NA, implying nonlinear data
 NOTE: HM81_flux.estimate is not NA, implying nonlinear data

```
head(fluxes)
```

	Plot	TIMESTAMP	HM81_AIC	HM81_flux.estimate		
1	10/14/2022_CRC_TR_309_71	2022-10-14 10:44:29	NA	NA		
2	10/14/2022_CRC_TR_311_72	2022-10-14 10:33:24	-345.05212	0.005212169		
3	10/14/2022_CRC_TR_312_73	2022-10-14 10:26:29	-39.85766	0.009751081		
4	10/14/2022_CRC_TR_313_74	2022-10-14 11:19:44	443.86657	0.008479056		
5	10/14/2022_CRC_TR_314_75	2022-10-14 10:06:29	-578.44904	0.004443127		
6	10/14/2022_CRC_TR_315_76	2022-10-14 10:13:19	-814.74178	0.003806261		
	HM81_p.value	HM81_r.squared	HM81_RMSE	lin_AIC	lin_flux.estimate	
1	NA	NA	NA	-2886.146	0.006374834	
2	6.617145e-124	0.6275136	0.1781514	-2771.974	0.005298659	
3	8.975891e-129	0.6618584	0.2323422	-2118.675	0.007647671	
4	5.370346e-201	0.6779560	0.3174565	-3855.737	0.007162640	
5	2.769750e-234	0.6957996	0.1750786	-4199.213	0.003651375	
6	3.788822e-135	0.6410119	0.1223071	-3318.155	0.003520039	
	lin_flux.std.error	lin_int.estimate	lin_int.std.error	lin_p.value		
1	1.379411e-05	14.94248	0.001427552	0		
2	1.618581e-05	14.40415	0.001768517	0		
3	2.807453e-05	14.96527	0.002905431	0		
4	1.006980e-05	14.33155	0.001565366	0		
5	9.015396e-06	17.78667	0.001557608	0		
6	1.073807e-05	14.81559	0.001235275	0		
	lin_r.squared	lin_RMSE	poly_AIC	poly_r.squared	poly_RMSE	rob_AIC
1	0.9974873	0.01665581	-2934.846	0.9977209	0.01589215	-2885.899
2	0.9947278	0.02119477	-2853.933	0.9954659	0.01969010	-2771.853
3	0.9928020	0.03389882	-2507.631	0.9965232	0.02360348	-2106.368
4	0.9984055	0.02233745	-4315.164	0.9991002	0.01680099	-3854.836
5	0.9945554	0.02342256	-5016.651	0.9978144	0.01485684	-4198.355
6	0.9944659	0.01518574	-3346.547	0.9947567	0.01480609	-3318.138
	rob_converged	rob_flux.estimate	rob_flux.std.error	rob_RMSE		
1	TRUE	0.006381624	1.446827e-05	0.01710613		
2	TRUE	0.005293765	1.638381e-05	0.02215159		
3	TRUE	0.007726051	2.603189e-05	0.03124424		
4	TRUE	0.007159559	1.003791e-05	0.02176887		
5	TRUE	0.003644753	9.219273e-06	0.02465547		
6	TRUE	0.003518718	1.146345e-05	0.01542092		
	TIMESTAMP_min	TIMESTAMP_max				
1	2022-10-14 10:43:00	2022-10-14 10:45:59				


```

2 2022-10-14 10:31:50 2022-10-14 10:34:59
3 2022-10-14 10:25:00 2022-10-14 10:27:59
4 2022-10-14 11:17:30 2022-10-14 11:21:59
5 2022-10-14 10:04:00 2022-10-14 10:08:59
6 2022-10-14 10:11:40 2022-10-14 10:14:59

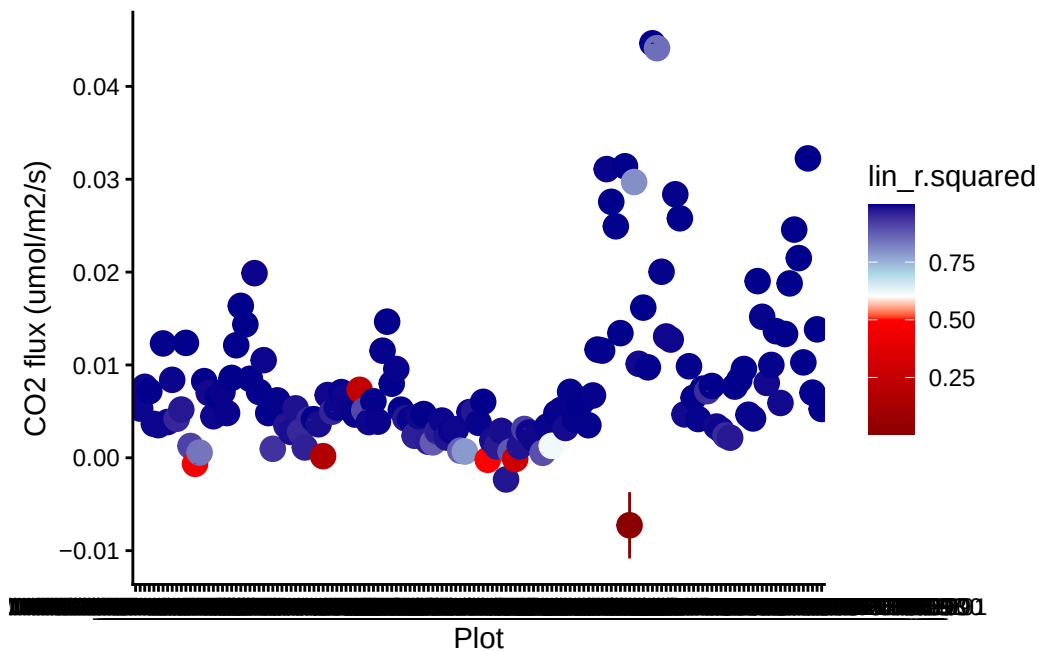
```

```

#merging fluxes with initial CO2 concentration
fluxes <- fluxes %>%
  left_join(df_CO2_in, by = "Plot")

#take a look at the fluxes we calculated with R2
ggplot(fluxes, aes(Plot, lin_flux.estimate, color = lin_r.squared)) +
  geom_point(size=4) + theme_classic() +
  geom_linerange(aes(ymin = lin_flux.estimate- lin_flux.std.error,
                    ymax = lin_flux.estimate+ lin_flux.std.error)) +
  scale_color_gradientn(colours = c("darkred", "red", "white", "lightblue", "darkblue"),
                        values = c(0, 0.5, 0.6, 0.7, 1)) +
  ylab("CO2 flux (umol/m2/s)")

```



```

#sort fluxes by R2
fluxes_good <- as.data.frame(subset(fluxes, lin_r.squared >= .90, select = Plot:TIMESTAMP_max))
fluxes_lowr2 <- as.data.frame(subset(fluxes, lin_r.squared < .90 & lin_flux.estimate < 1, select = Plot:TIMESTAMP_max))

```

```

fluxes_weird <- as.data.frame(subset(fluxes, lin_r.squared < .90, select = Plot:TIMESTAMP_ma

#find fluxes that represent the first and fourth quartile of the data
low <- quantile(fluxes$lin_flux.estimate, 0.05)
high <- quantile(fluxes$lin_flux.estimate, 0.95)

fluxes_low <- as.data.frame(subset(fluxes, lin_flux.estimate <= low, select = Plot:TIMESTAMP
fluxes_high <- as.data.frame(subset(fluxes, lin_flux.estimate >= high, select = Plot:TIMESTAI

tails <- rbind(fluxes_low, fluxes_high)

```

11 check the fluxes for which I changed the start time due to lags or high values (recorded in notes of metadata)

```

# grab trim limits for each overridden Plot
trim_limits <- alldat_trimmed %>%
  filter(Plot %in% time_overrides$Plot) %>%
  group_by(Plot) %>%
  summarise(
    t_start_trim = min(TIMESTAMP),
    t_end_trim   = max(TIMESTAMP) )

# use full alldat_matched + trim limits to classify each point
r_all <- alldat_matched %>%
  filter(Plot %in% time_overrides$Plot) %>%
  left_join(trim_limits, by = "Plot") %>%
  group_by(Plot) %>%
  mutate(
    time_sec = as.numeric(difftime(TIMESTAMP, min(TIMESTAMP), units = "secs")),
    point_status = case_when(
      TIMESTAMP < t_start_trim ~ "Discarded (deadband)",
      TIMESTAMP > t_end_trim   ~ "Discarded (stop trim)",
      TRUE                     ~ "Used"   )) %>%
  ungroup()

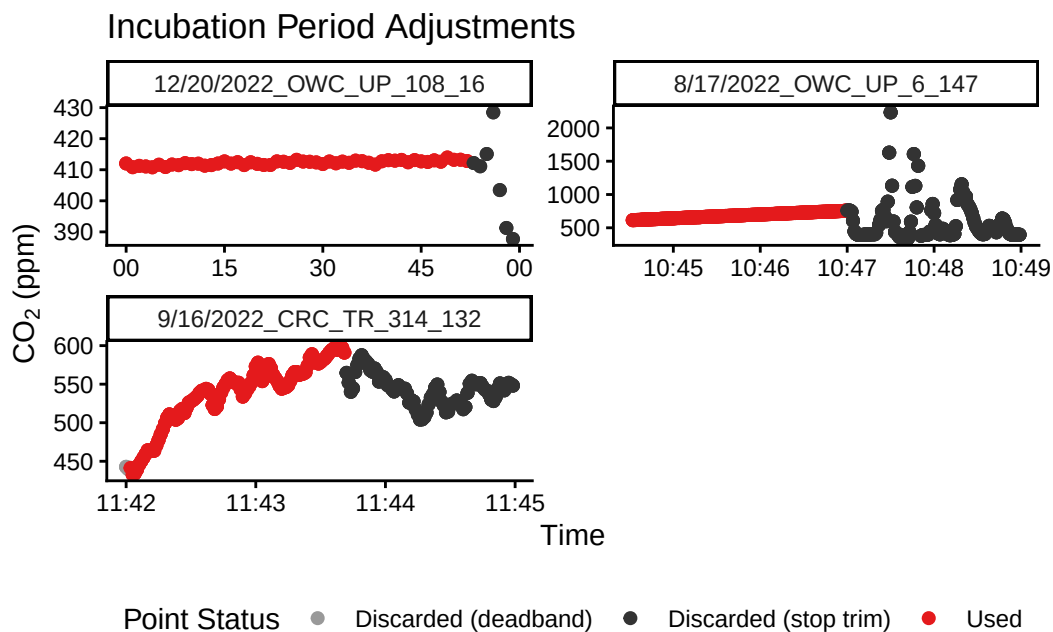
# plot all overridden plots
ggplot(r_all, aes(x = TIMESTAMP, y = CO2, color = point_status)) +
  geom_point(size = 1.8) +
  facet_wrap(~ Plot, scales = "free", ncol = 2) +

```

```

scale_color_manual(
  values = c(
    "Used" = "#E41A1C",
    "Discarded (deadband)" = "grey60",
    "Discarded (stop trim)" = "grey20"
  )
) +
theme_classic() +
ylab(expression(paste(CO[2], " (ppm)"))) +
xlab("Time") +
labs(
  title = "Incubation Period Adjustments",
  color = "Point Status") +
theme(legend.position = "bottom")

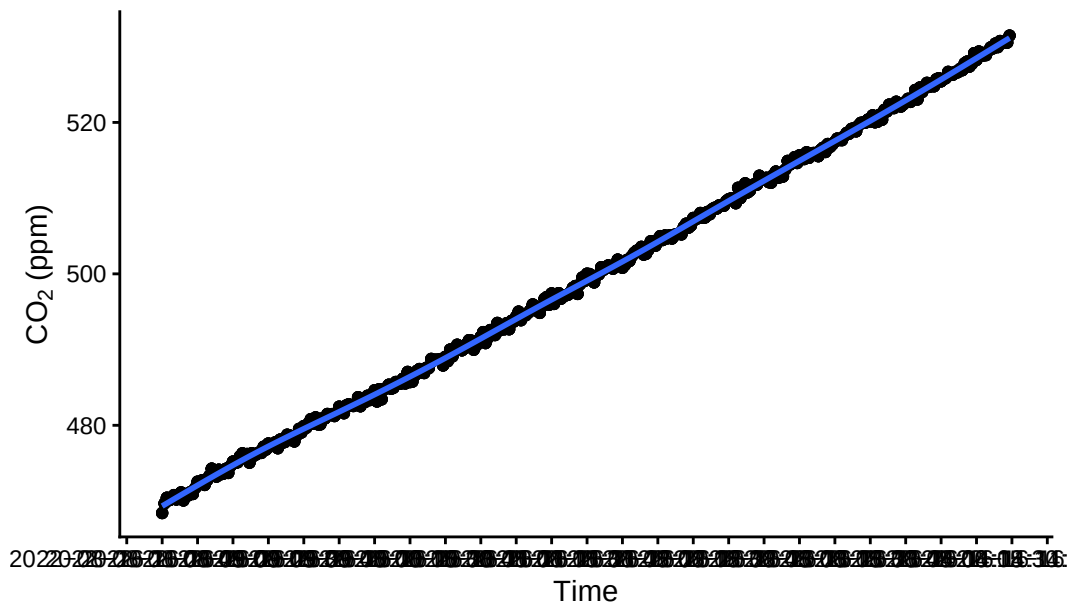
```



12 Spot check a few fluxes with $R^2 > 0.9$

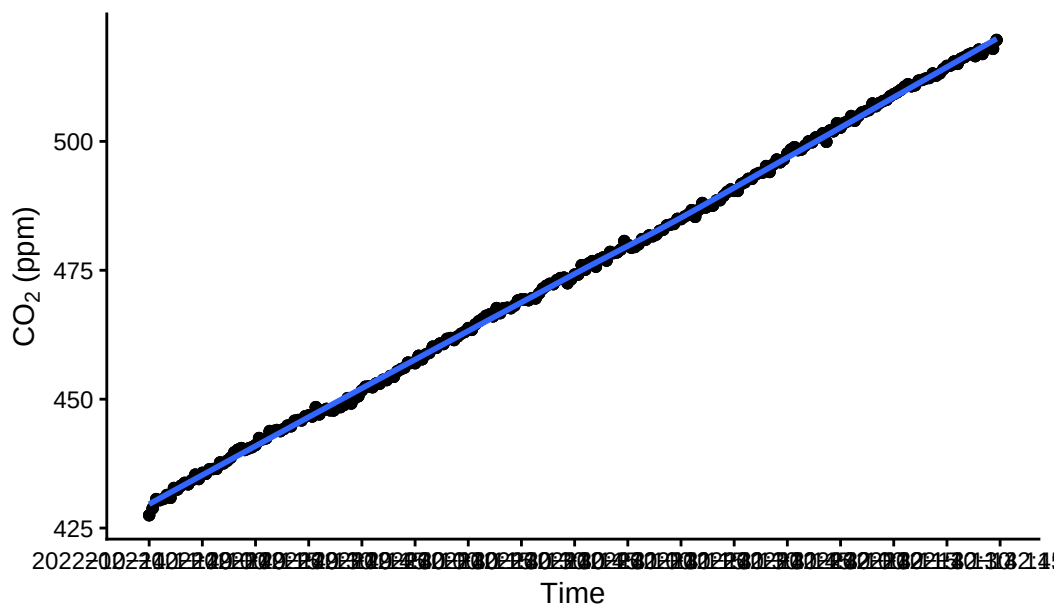
`geom\_smooth()` using method = 'gam' and formula = 'y ~ s(x, bs = "cs")'

8/16/2022_CRC_TR_314_151

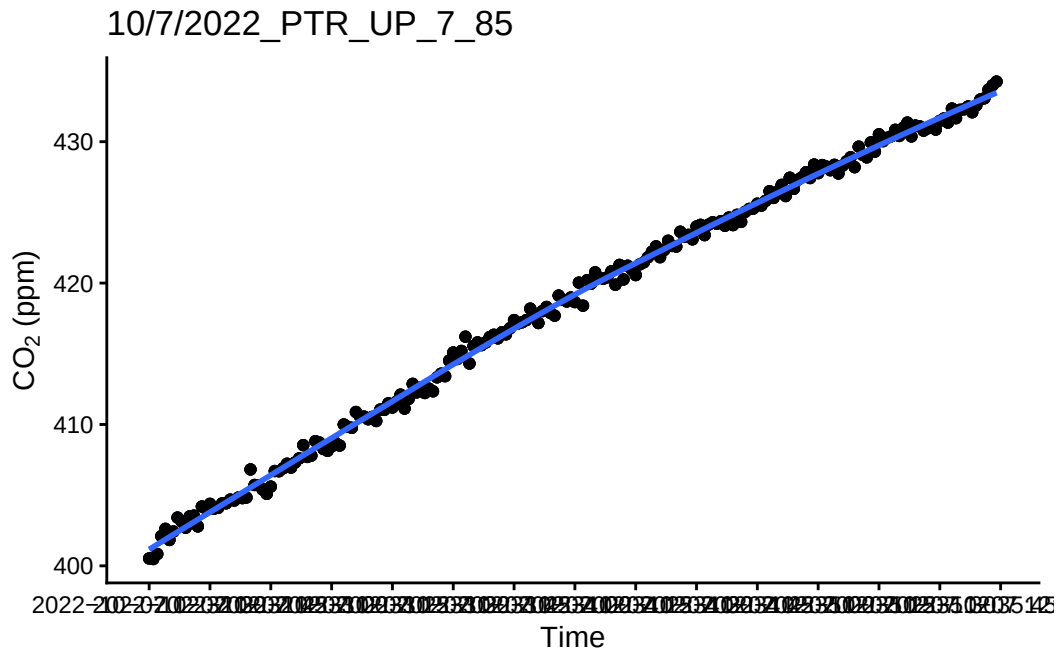


``geom_smooth()` using method = 'loess' and formula = 'y ~ x'`

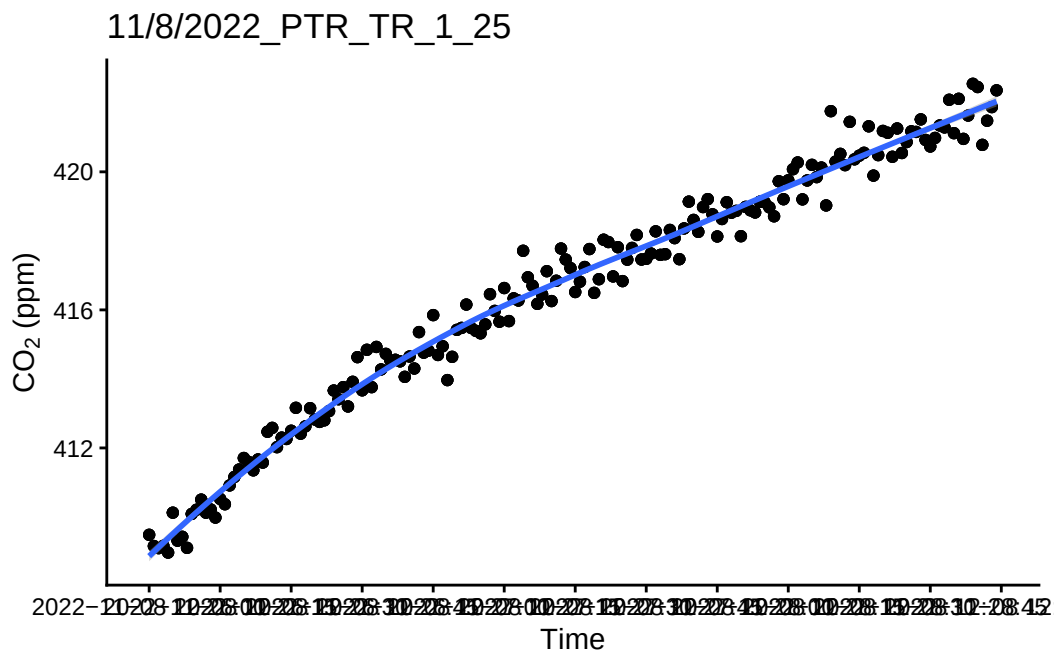
10/14/2022_CRC_TR_316_77



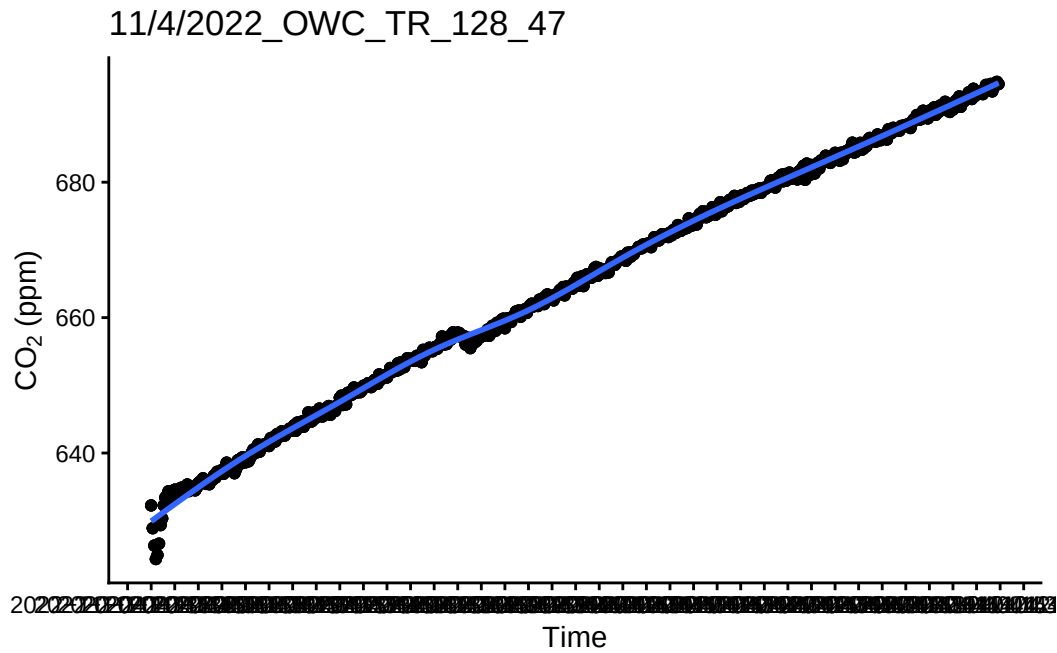
``geom_smooth()` using method = 'loess' and formula = 'y ~ x'`



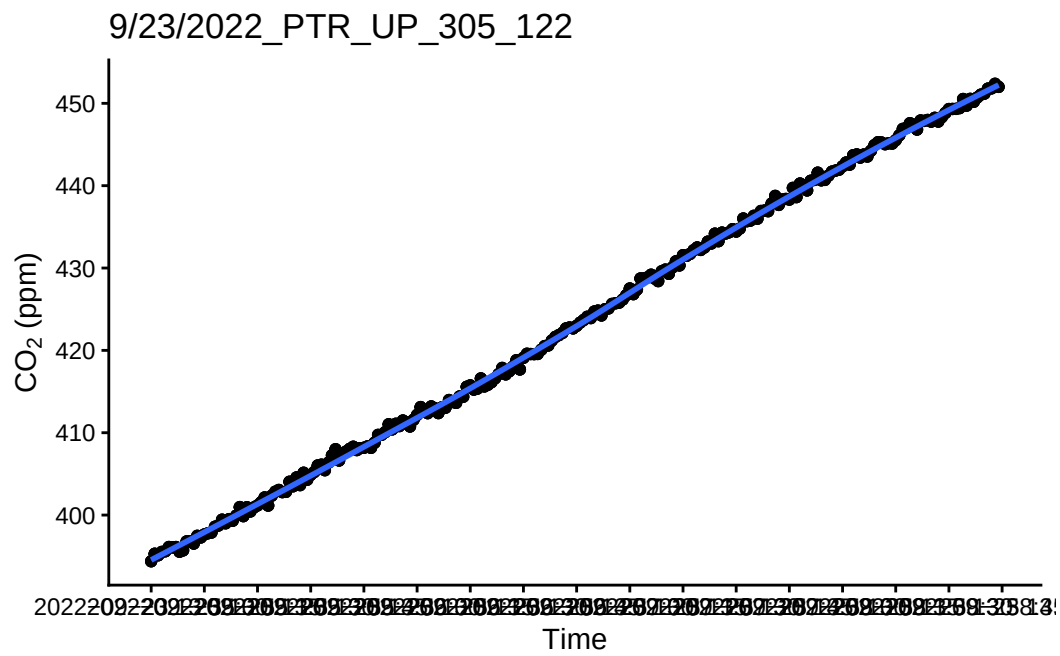
``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'



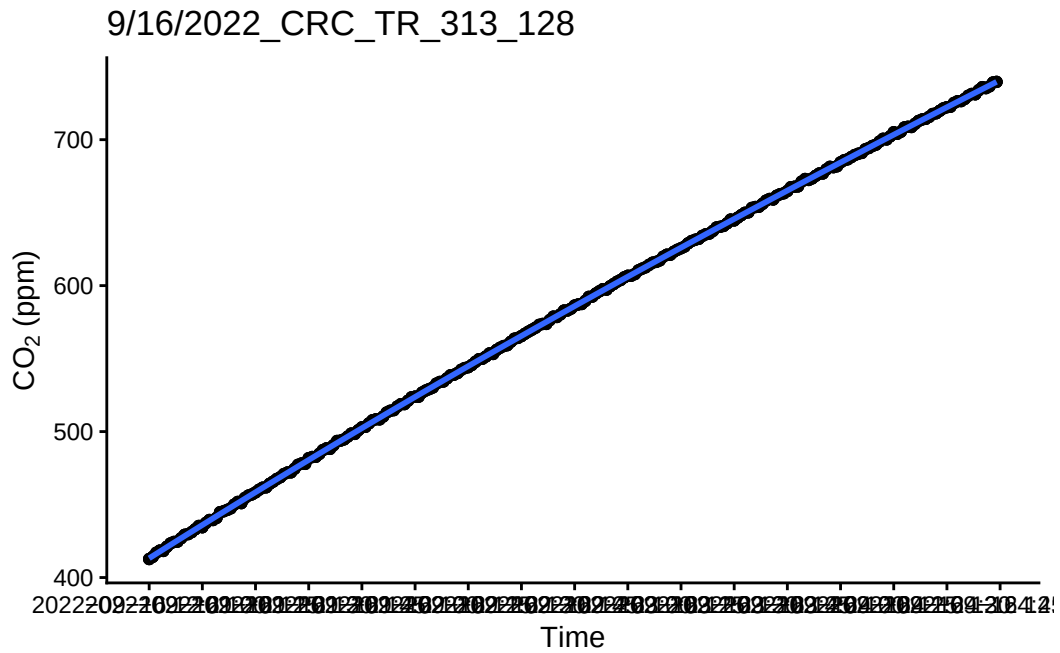
``geom_smooth()`` using method = 'gam' and formula = 'y ~ s(x, bs = "cs")'



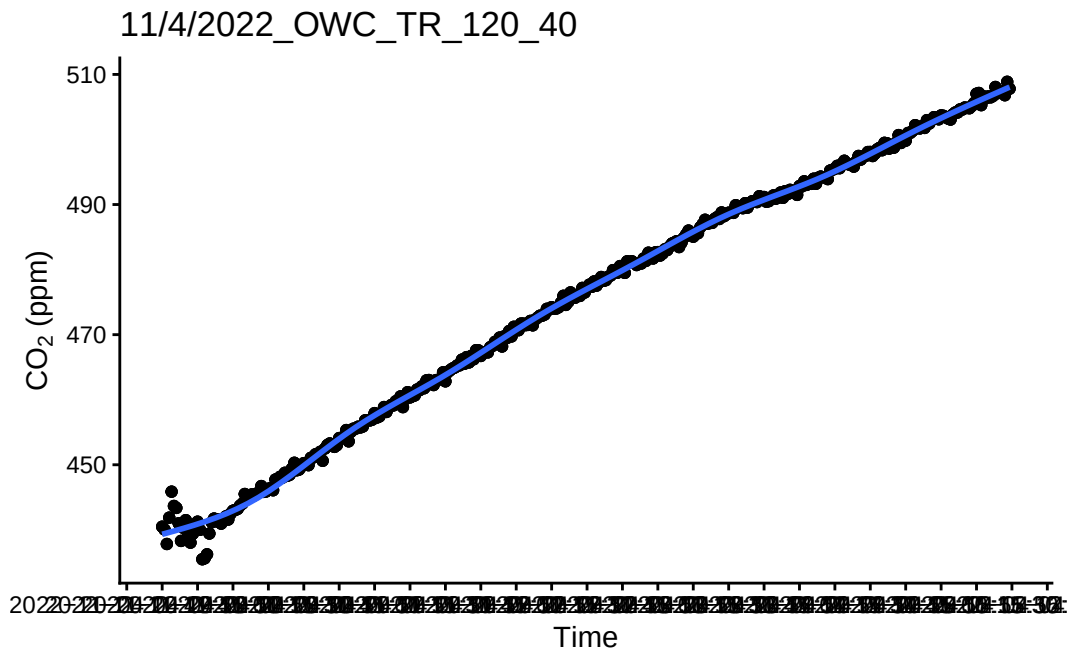
``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'



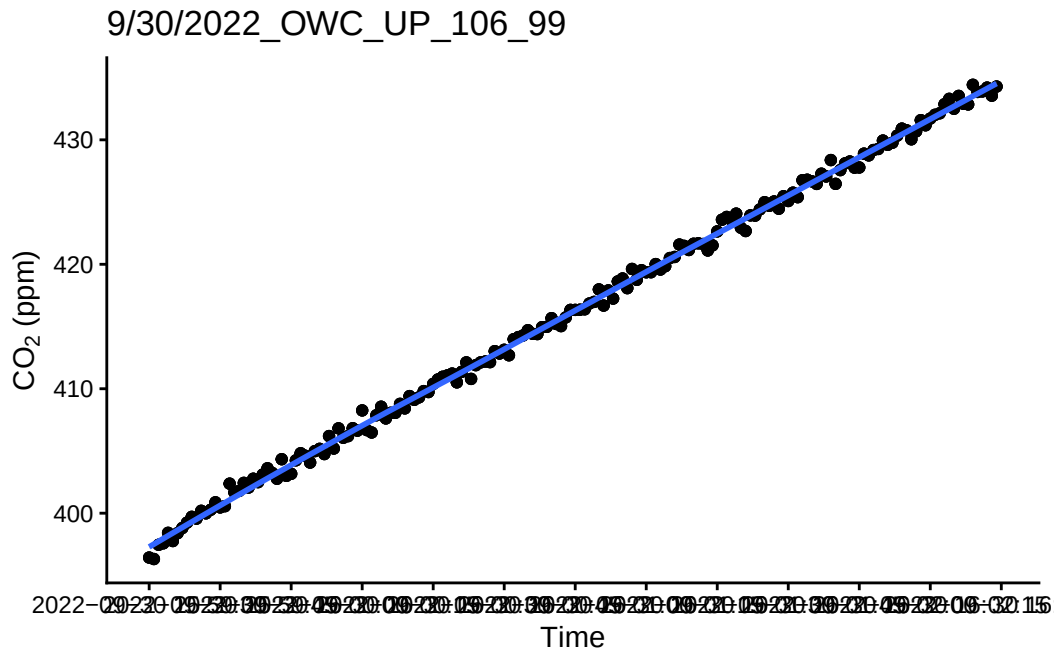
``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'



``geom_smooth()`` using method = 'gam' and formula = 'y ~ s(x, bs = "cs")'

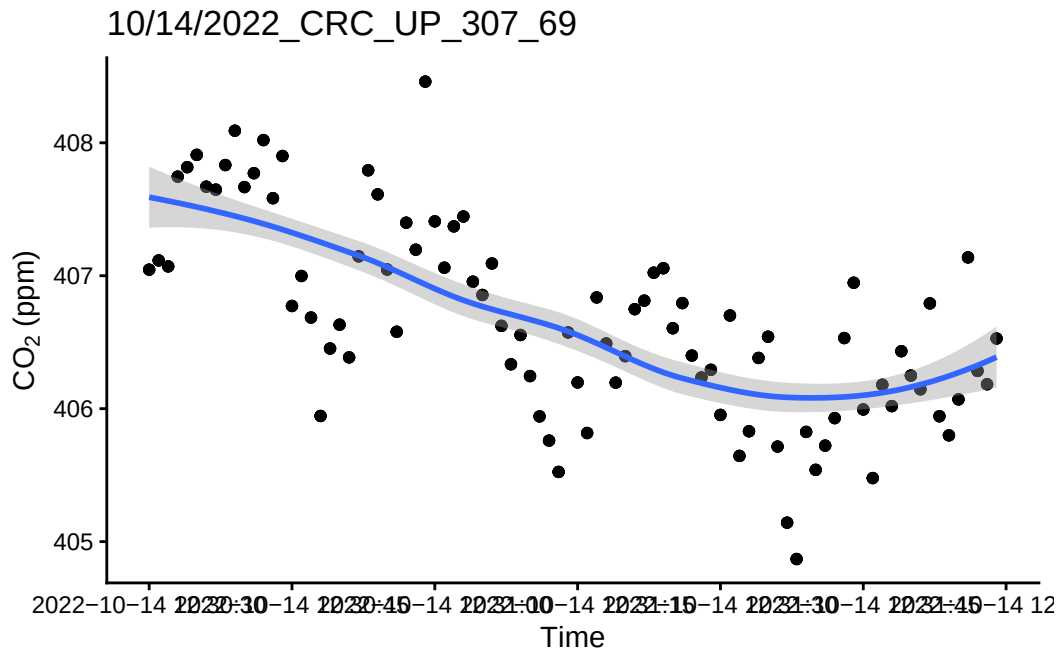


``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'

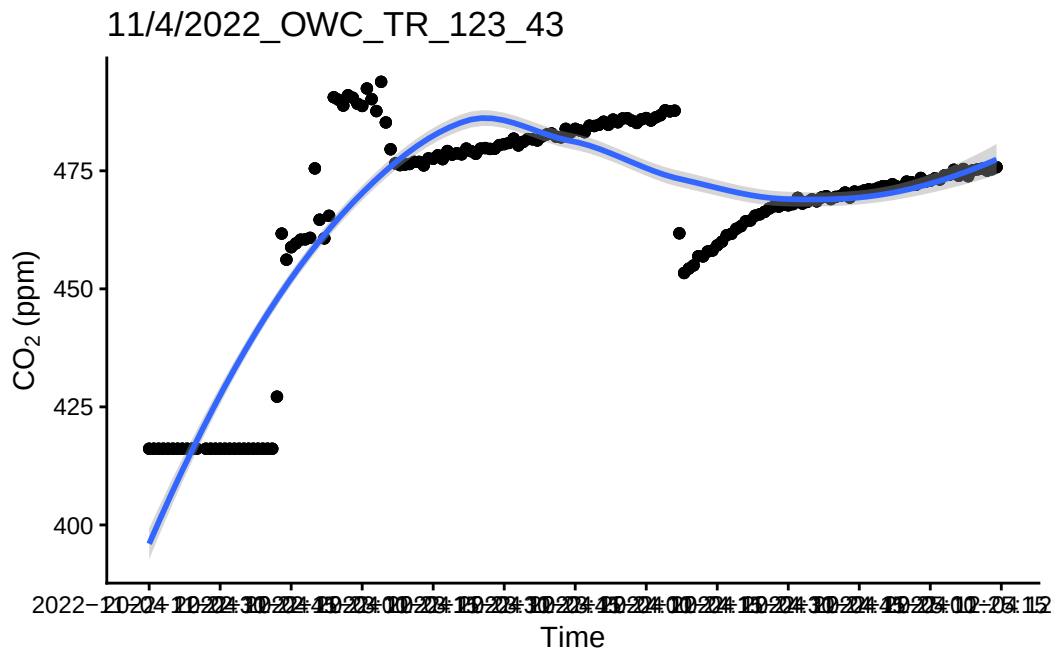


13 Take a look at the fluxes < 0.9 and a flux estimate below 1

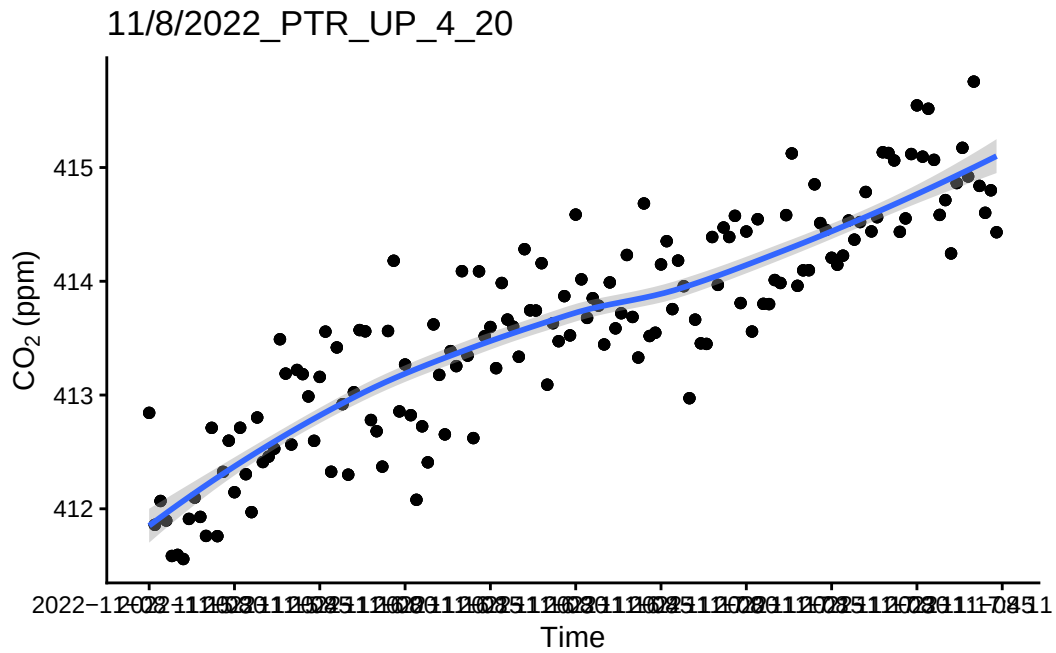
``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'



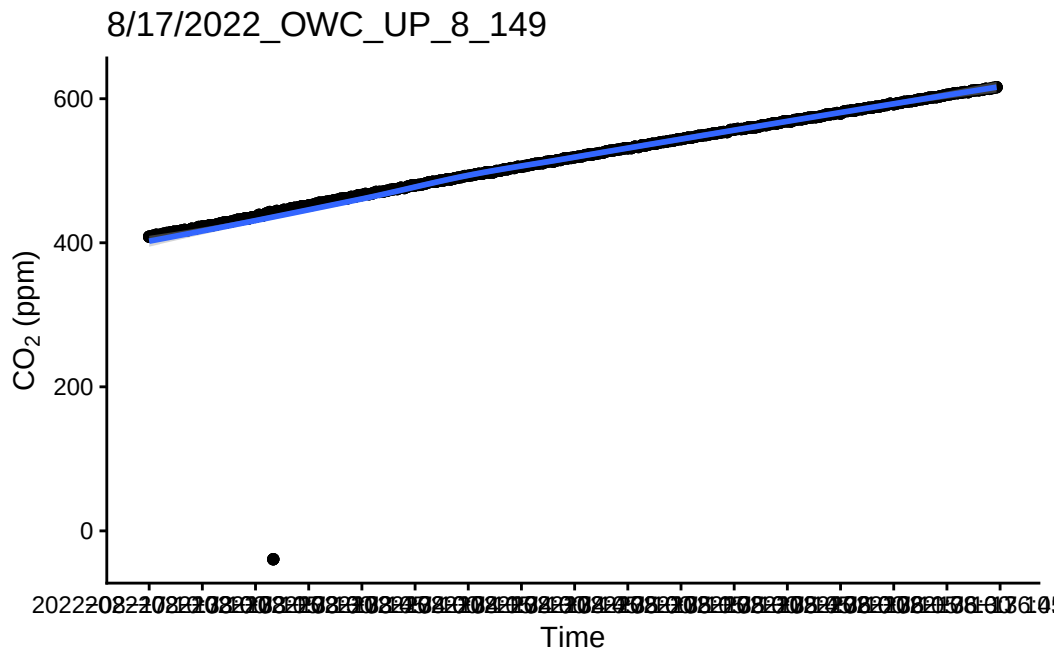
``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'



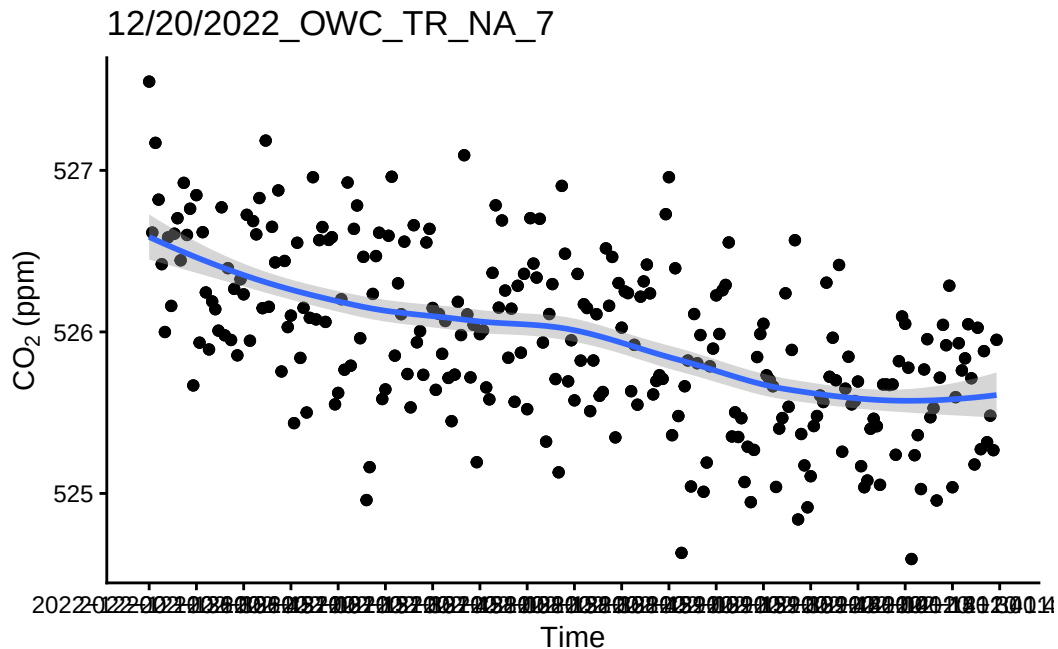
``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'



``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'



``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'



14 Take a look at the bad fluxes <0.9 and have a high flux estimate

15 See if we want to remove these fluxes or if we want to try and recalculate these ones?

```
### pull out fluxes with R2 < 0.9 and put them in a list
bad_plots <- unique(fluxes_weird$Plot)

# take a random sample (FALSE so we don't get the same plot twice)
# set.seed(...) to guarantee reproducibility
bad_sample <- sample(bad_plots, 5, replace=FALSE)

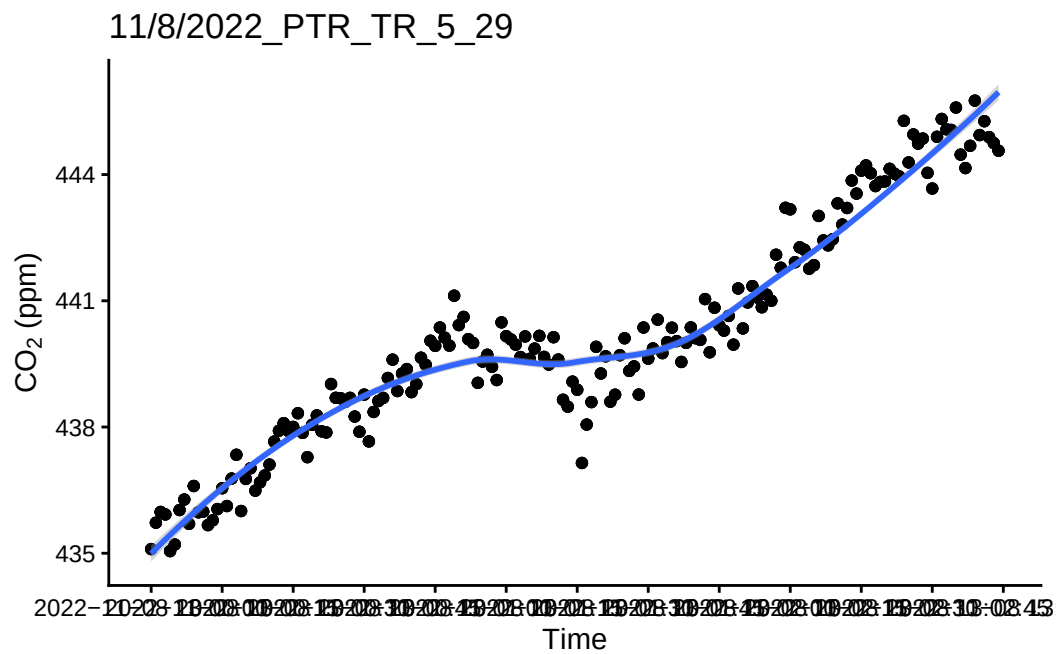
#Loop to run five random plots subset above
for(P in bad_sample){
  r1 <- subset(alldat_trimmed, Plot == P & !is.na(metadata_row))
  p1 <- ggplot(r1, aes(TIMESTAMP, CO2)) + geom_point() + theme_classic() +
    scale_x_datetime(breaks = "15 sec") +
    ylab(expression(paste( CO[2], " (ppm)"))) +
```

```

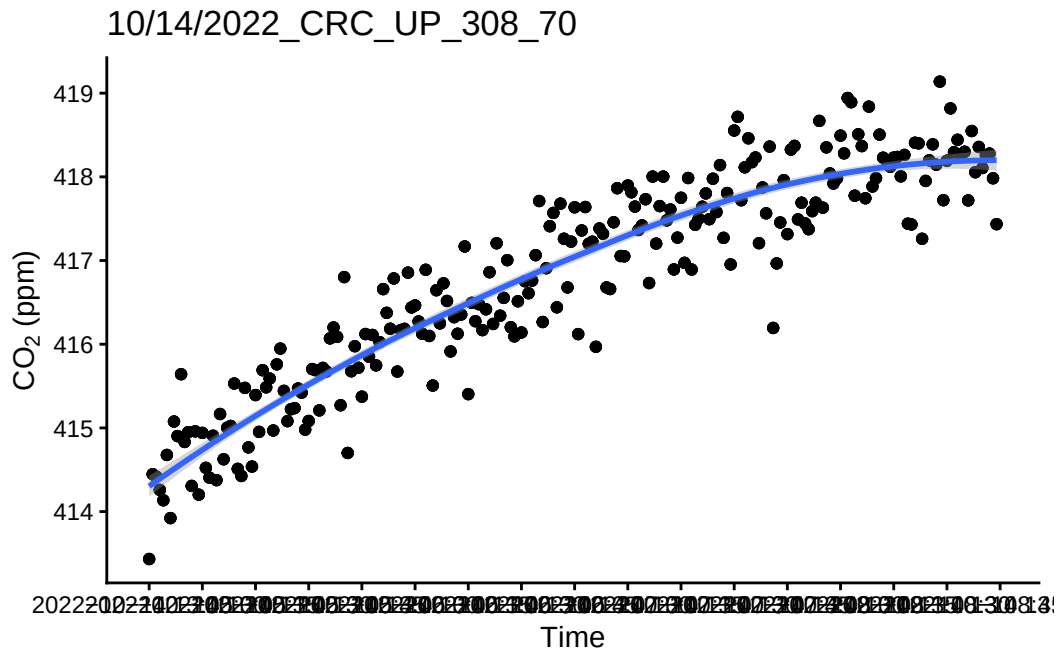
xlab("Time") + labs(title = P) +
geom_smooth()
print(p1)
}

```

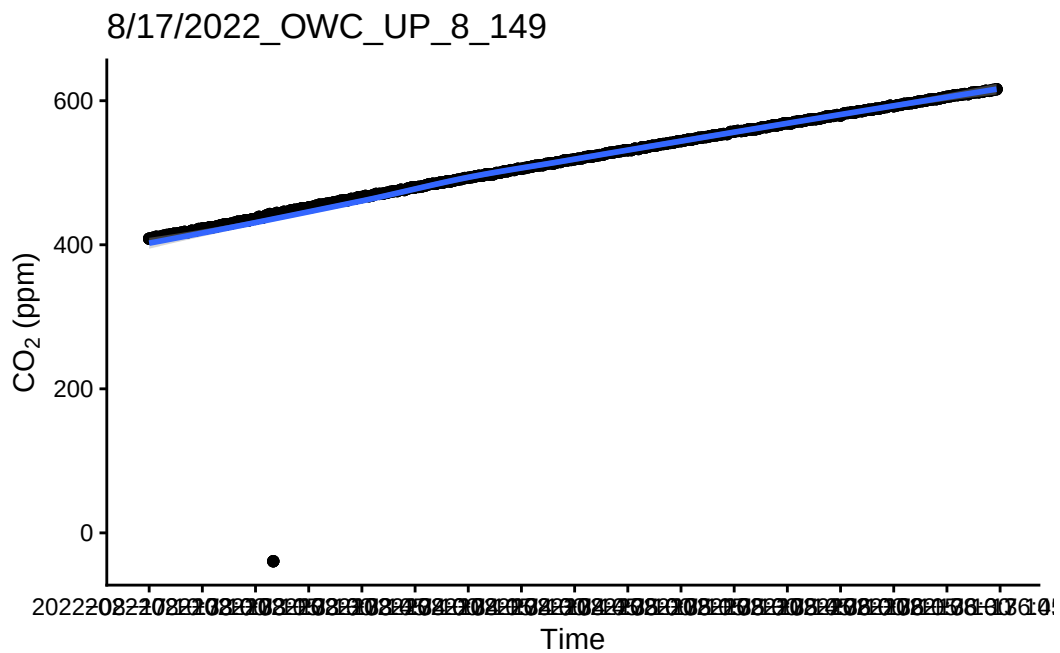
`geom\_smooth()` using method = 'loess' and formula = 'y ~ x'



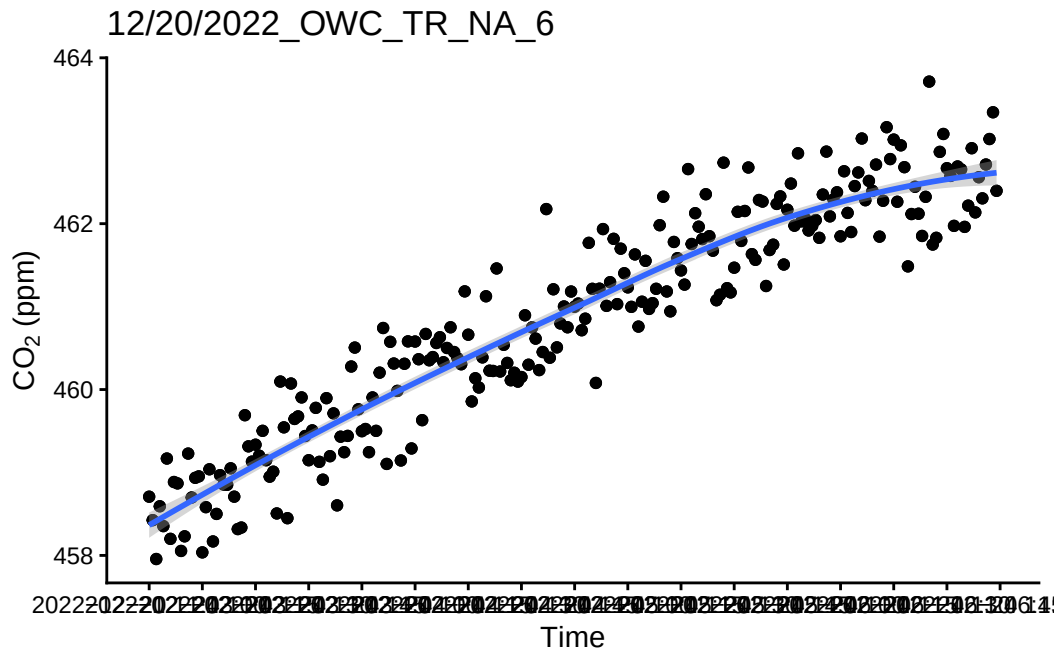
`geom\_smooth()` using method = 'loess' and formula = 'y ~ x'



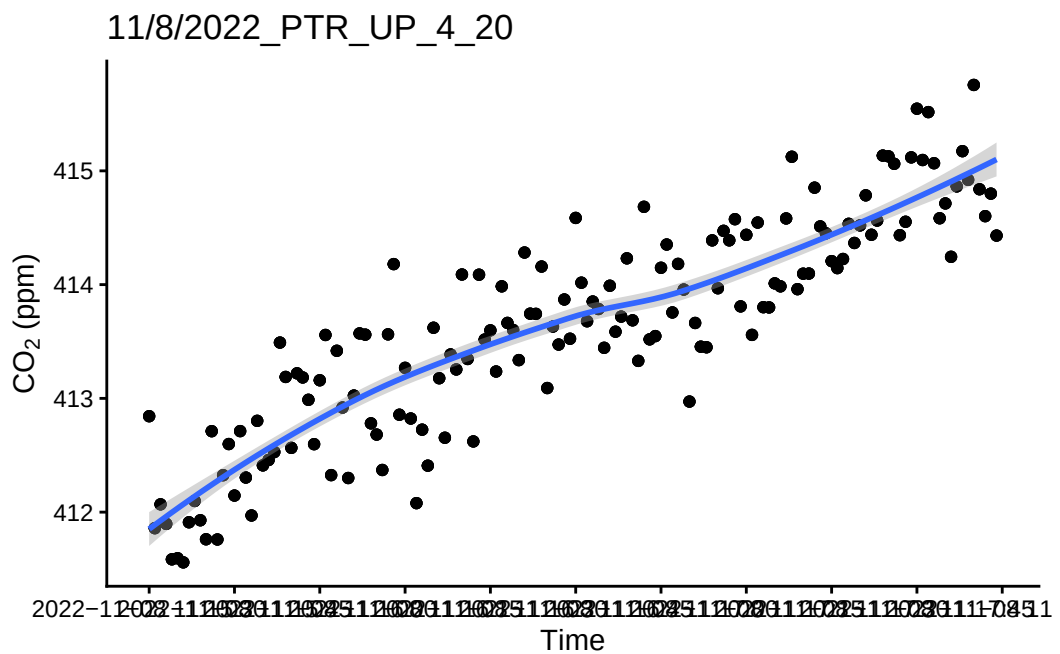
``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'



``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'



``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'



```
#####
```

16 Look at the really high or really low ones to see if there is a timing issue or something

```
### pull out fluxes in the 5% and 95% tails to look at really high and really low
tail_plots <- unique(tails$Plot)

# take a random sample (FALSE so we don't get the same plot twice)
# set.seed(...) to guarantee reproducibility
tail_sample <- sample(tail_plots, 2, replace=FALSE)

#Loop to run five random plots subset above
for(P in tail_sample){
  r1 <- subset(alldat_trimmed, Plot == P & !is.na(metadata_row))
  p1 <- ggplot(r1, aes(TIMESTAMP, CO2)) + geom_point() + theme_classic() +
    scale_x_datetime(breaks = "15 sec") +
    ylab(expression(paste( CO [2], " (ppm)"))) +
    xlab("Time") + labs(title = P) +
    geom_smooth()
  print(p1)
}
```

`geom\_smooth()` using method = 'loess' and formula = 'y ~ x'

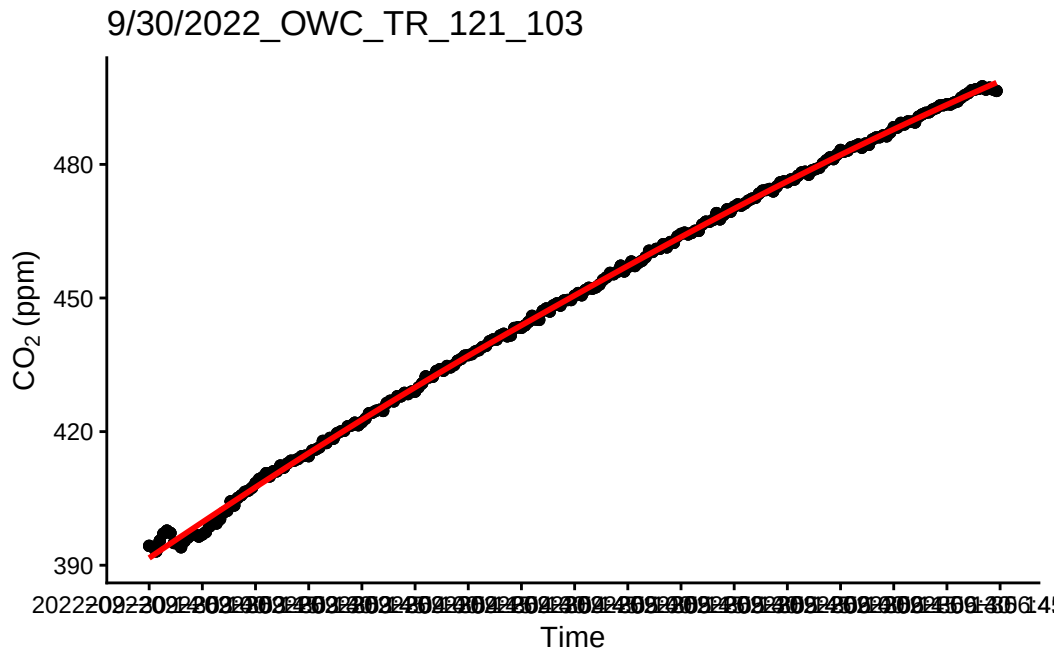
The figure is a scatter plot with a trend line and a confidence interval. The y-axis is labeled 'CO₂ (ppm)' and ranges from 450 to 600. The x-axis is labeled 'Time' and shows dates from 2022-09-16 to 2024-09-16. Black dots represent individual CO₂ measurements. A solid blue line represents the fitted trend, and a grey shaded area around it represents the confidence interval. The data shows a clear upward trend in CO₂ levels over time.


```
#would be nice to put on the plot the flux estimate and if it is a high or low or something  
#####
```

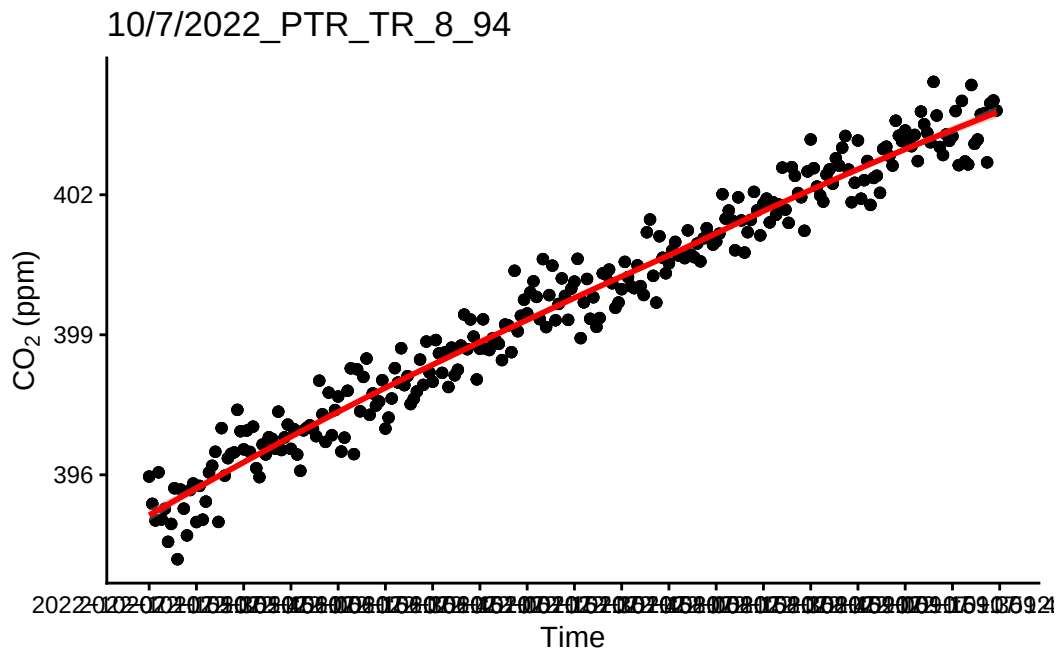
17 Check all fluxes

```
##Put all fluxes together  
#fluxes_all <- as.data.frame(subset(fluxes, select = Plot:TIMESTAMP_max))  
  
all_plots <- unique(fluxes$Plot)  
  
# take a random sample (FALSE so we don't get the same plot twice)  
# set.seed(...) to guarantee reproducibility  
all_sample <- sample(all_plots, 5, replace=FALSE)  
  
#Loop to run five random plots subset above  
for(P in all_sample){  
  r1 <- subset(alldat_trimmed, Plot== P & !is.na(metadat_row))  
  p1 <- ggplot(r1, aes(TIMESTAMP, CO2)) + geom_point() + theme_classic() +  
    scale_x_datetime(breaks = "15 sec") +  
    ylab(expression(paste( CO [2], " (ppm)"))) +  
    xlab("Time") + labs(title = P) +  
    geom_smooth(color="red")  
  print(p1)  
}
```

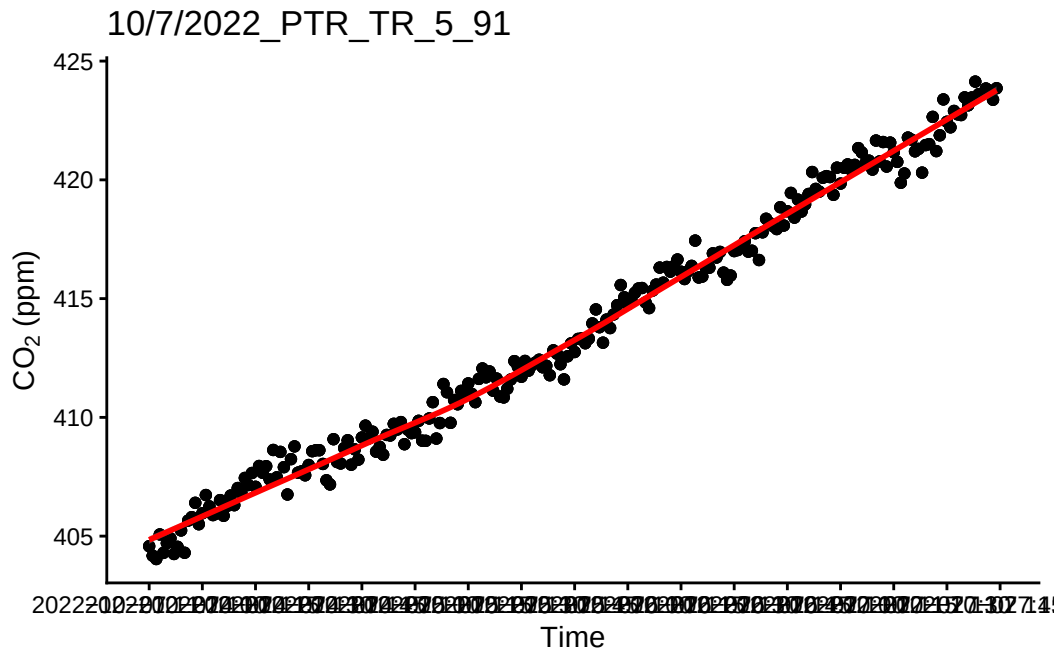
`geom\_smooth()` using method = 'loess' and formula = 'y ~ x'



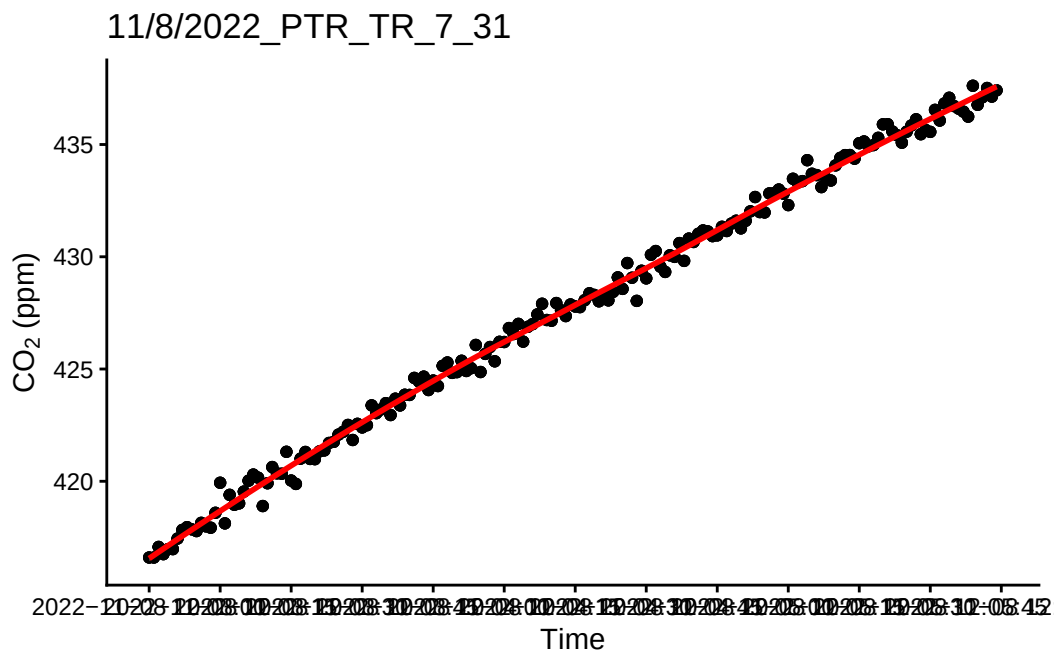
``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'



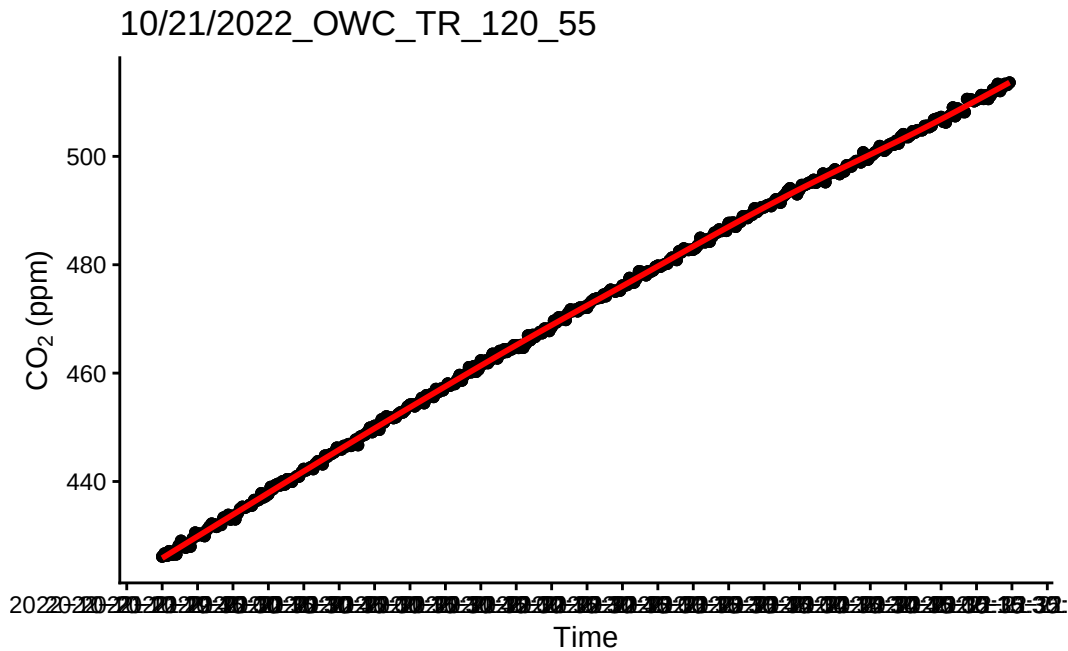
``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'



``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'



``geom_smooth()`` using method = 'gam' and formula = 'y ~ s(x, bs = "cs")'



#####

18 QAQC - Flags

```
fluxes_flagged <- fluxes %>%
  mutate(
    # model fit flags
    CO2_lin_pass = (lin_r.squared >= 0.85 | lin_RMSE < 5) & !is.na(lin_r.squared),
    CO2_HM81_pass = (HM81_r.squared >= 0.85 | HM81_RMSE < 5) & !is.na(HM81_r.squared),
    CO2_poly_pass = (poly_r.squared >= 0.85 | poly_RMSE < 5) & !is.na(poly_r.squared),
    # negative flux flag
    CO2_neg_flux = lin_flux.estimate < 0 | is.na(lin_flux.estimate),
    # rejection flags
    CO2_rejected_model = !(CO2_lin_pass | CO2_HM81_pass | CO2_poly_pass),
    CO2_rejected_negflux = CO2_neg_flux,
    CO2_flux_rejected = CO2_rejected_model | CO2_rejected_negflux
  )

# percentage of problems
r2_summary <- fluxes_flagged %>%
```

```

summarise(
  CO2_total_obs      = n(),
  CO2_n_rejected_model = sum(CO2_rejected_model, na.rm = TRUE),
  CO2_n_rejected_negflux = sum(CO2_rejected_negflux, na.rm = TRUE),
  CO2_n_rejected_both  = sum(CO2_rejected_model & CO2_rejected_negflux, na.rm = TRUE),
  CO2_n_rejected_total = sum(CO2_flux_rejected, na.rm = TRUE),
  CO2_pct_rejected_total = 100 * CO2_n_rejected_total / CO2_total_obs,
  CO2_n_lin_pass       = sum(CO2_lin_pass, na.rm = TRUE),
  CO2_n_HM81_pass      = sum(CO2_HM81_pass, na.rm = TRUE),
  CO2_n_poly_pass      = sum(CO2_poly_pass, na.rm = TRUE)
)

# formatted table for PDF
r2_summary %>%
  tidyr::pivot_longer(everything(), names_to = "Metric", values_to = "Value") %>%
  mutate(
    Metric = recode(Metric,
      CO2_total_obs      = "Total observations",
      CO2_n_rejected_model = "Rejected - poor model fit (n)",
      CO2_n_rejected_negflux = "Rejected - negative CO2 flux (n)",
      CO2_n_rejected_both  = "Rejected - both reasons (n)",
      CO2_n_rejected_total = "Rejected - total (n)",
      CO2_pct_rejected_total = "Rejected - total (%)",
      CO2_n_lin_pass       = "Passed - Linear model (n)",
      CO2_n_HM81_pass      = "Passed - HM81 model (n)",
      CO2_n_poly_pass      = "Passed - Polynomial model (n)"
    ),
    Value = ifelse(Metric == "Rejected - total (%)",
      round(Value, 1),
      as.integer(Value)) ) %>%
  kable(
    caption = "QA/QC CO2 Flux Summary",
    align   = c("l", "r"),
    booktabs = TRUE,
    format   = "latex") %>%
  kable_styling(
    latex_options = c("striped", "hold_position"),
    full_width     = FALSE ) %>%
  pack_rows("Rejections", 2, 6) %>%
  pack_rows("Model performance", 7, 9)

```

Table 1: QA/QC CO2 Flux Summary

Metric	Value
Total observations	151.0
Rejections	
Rejected — poor model fit (n)	1.0
Rejected — negative CO2 flux (n)	5.0
Rejected — both reasons (n)	1.0
Rejected — total (n)	5.0
Rejected — total (%)	3.3
Model performance	
Passed — Linear model (n)	150.0
Passed — HM81 model (n)	117.0
Passed — Polynomial model (n)	150.0

```
# R2 vs RMSE diagnostic plot
# pivot_longer keeps R2/RMSE columns and uses CO2*_pass as the pass indicator
set.seed(42)
plot_data <- fluxes_flagged %>%
  slice_sample(n = min(15000, nrow(.))) %>%
  select(
    lin_r.squared, lin_RMSE, CO2_lin_pass,
    HM81_r.squared, HM81_RMSE, CO2_HM81_pass,
    poly_r.squared, poly_RMSE, CO2_poly_pass,
    CO2_rejected_negflux
  ) %>%
  # rename pass columns to match model prefix before pivoting
  rename(
    lin_pass = CO2_lin_pass,
    HM81_pass = CO2_HM81_pass,
    poly_pass = CO2_poly_pass
  ) %>%
  pivot_longer(
    cols = -CO2_rejected_negflux,
    names_to = c("model", ".value"),
    names_pattern = "(lin|HM81|poly)_(.*)"
  ) %>%
  rename(r2 = r.squared) %>%
  mutate(
    model = recode(model, lin = "Linear", HM81 = "HM81", poly = "Polynomial"),
```

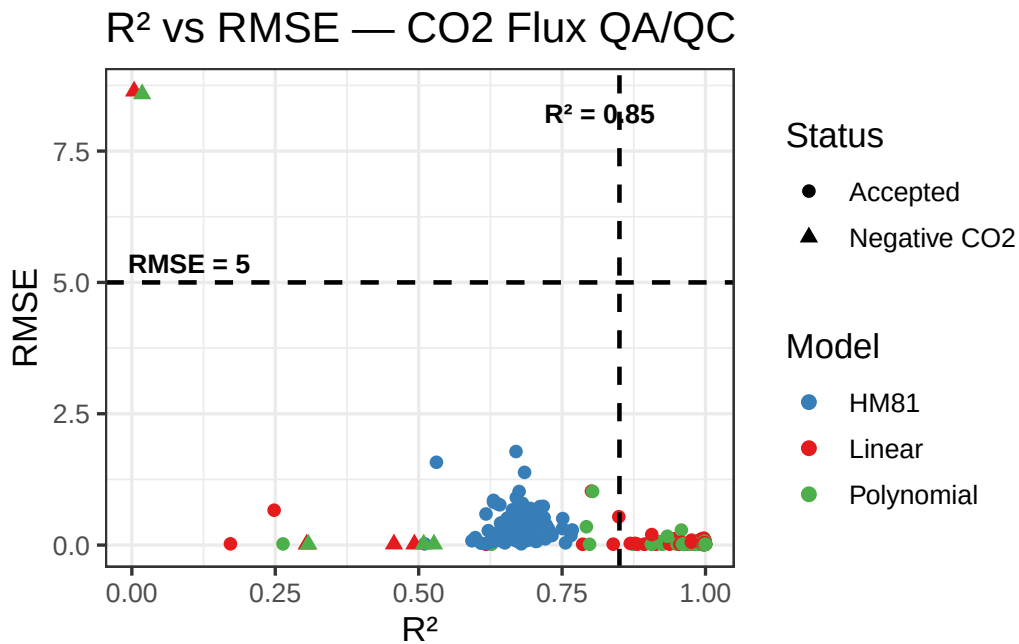
```

    status = case_when(
      CO2_rejected_negflux ~ "Negative CO2",
      pass                  ~ "Accepted",
      TRUE                 ~ "Rejected"
    )
  ) %>%
  filter(!is.na(r2), !is.na(RMSE))

#graphing
QAQC_plot <- ggplot(plot_data, aes(x = r2, y = RMSE, color = model, shape = status)) +
  geom_point(size = 2) +
  geom_vline(xintercept = 0.85, linetype = "dashed", color = "black", linewidth = 0.8) +
  geom_hline(yintercept = 5, linetype = "dashed", color = "black", linewidth = 0.8) +
  annotate("text", x = 0.7, y = max(plot_data$RMSE, na.rm = TRUE) * 0.95,
    label = "R2 = 0.85", hjust = -0.1, size = 3.5, fontface = "bold") +
  annotate("text", x = 0.1, y = 5,
    label = "RMSE = 5", vjust = -0.5, size = 3.5, fontface = "bold") +
  scale_color_manual(values = c(
    "Linear"      = "#E41A1C",
    "HM81"       = "#377EB8",
    "Polynomial" = "#4DAF4A"
  )) +
  scale_shape_manual(values = c(
    "Accepted"    = 16,
    "Rejected"    = 4,
    "Negative CO2" = 17 )) +
  labs(
    title = "R2 vs RMSE - CO2 Flux QA/QC",
    x     = "R2",
    y     = "RMSE",
    color = "Model",
    shape = "Status" ) +
  theme_bw(base_size = 13) +
  theme(legend.position = "right")

print(QAQC_plot)

```



19 Adding manual flag to reject any fluxes that do not fit criteria

```
# adding flags to the ones that are crazy off based on visual checks
manual_reject <- c(
  "8/17/2022_OWC_UP_7_148",
  "8/17/2022_OWC_UP_6_147",
  "11/4/2022_OWC_TR_123_43"
)

fluxes_flagged <- fluxes_flagged %>%
  mutate(
    CO2_rejected_manual = Plot %in% manual_reject,
    CO2_flux_rejected = CO2_rejected_model | CO2_rejected_negflux | CO2_rejected_manual)

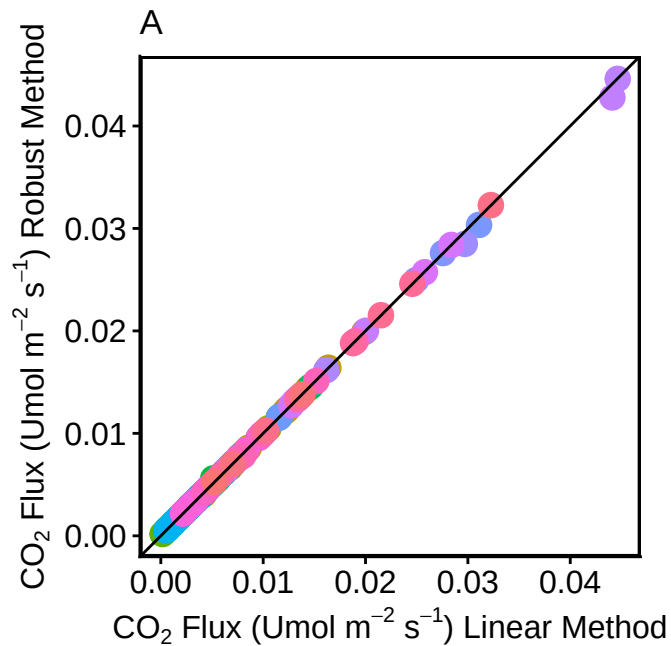
# fluxes_flagged %>%
#   filter(CO2_rejected_manual) %>%
#   select(Plot, lin_r.squared, lin_RMSE, CO2_rejected_model, CO2_rejected_negCO2, CO2_rejected_manual)
```


20 Do some visualization of QAQC plots to assess fluxes

```
allflux_filtered <- fluxes_flagged %>% filter(!CO2_flux_rejected)

p1 <- ggplot(allflux_filtered, aes(lin_flux.estimate, rob_flux.estimate, color = Plot)) +
  geom_point(size=4) + geom_abline() +
  theme_classic() +
  labs(title = "A") +
  ylab(expression(paste(CO [2], " Flux (Umol m-2 s-1) Robust Method"))) +
  xlab(expression(paste(CO [2], " Flux (Umol m-2 s-1) Linear Method"))) +
  guides(color = guide_legend(title = "Plot")) +
  theme(legend.position="none") +
  theme(axis.title.x = element_text(size=12), axis.text = element_text(size=12),
        axis.title.y = element_text(size=12), legend.text=element_text(size=12),
        panel.border = element_rect(colour = "black", fill=NA, linewidth = 1),
        aspect.ratio = 1)

p1
```



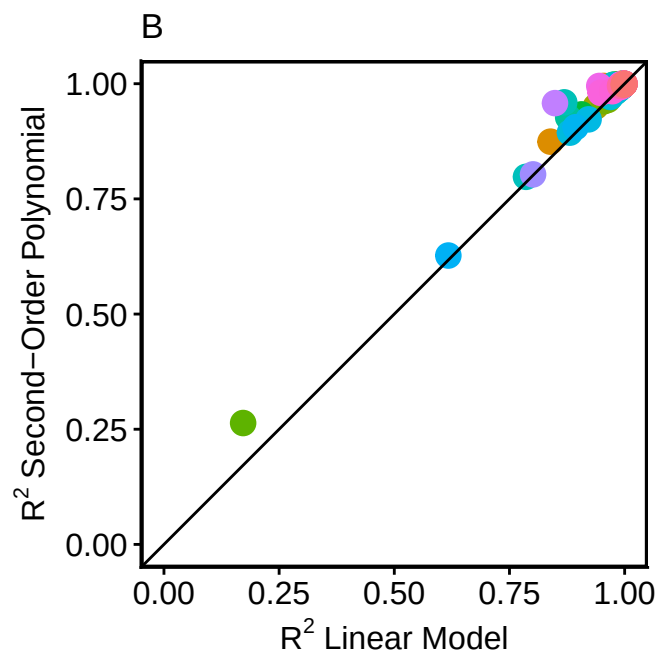
```
p2 <- ggplot(allflux_filtered, aes(lin_r.squared, poly_r.squared, color = Plot)) +
  geom_point(size=4) + geom_abline() +
```

```

theme_classic() +
xlim(0,1) + ylim(0,1) +
labs(title = "B") +
ylab(expression(paste("R"2* " Second-Order Polynomial")))) +
xlab(expression(paste("R"2* " Linear Model")))) +
guides(color = guide_legend(title = "Plot_ID")) +
theme(legend.position="none") +
theme(axis.title.x = element_text(size=12), axis.text = element_text(size=12),
      axis.title.y = element_text(size=12), legend.text=element_text(size=12),
      panel.border = element_rect(colour = "black", fill=NA, linewidth =1),
      aspect.ratio = 1)

```

p2



```

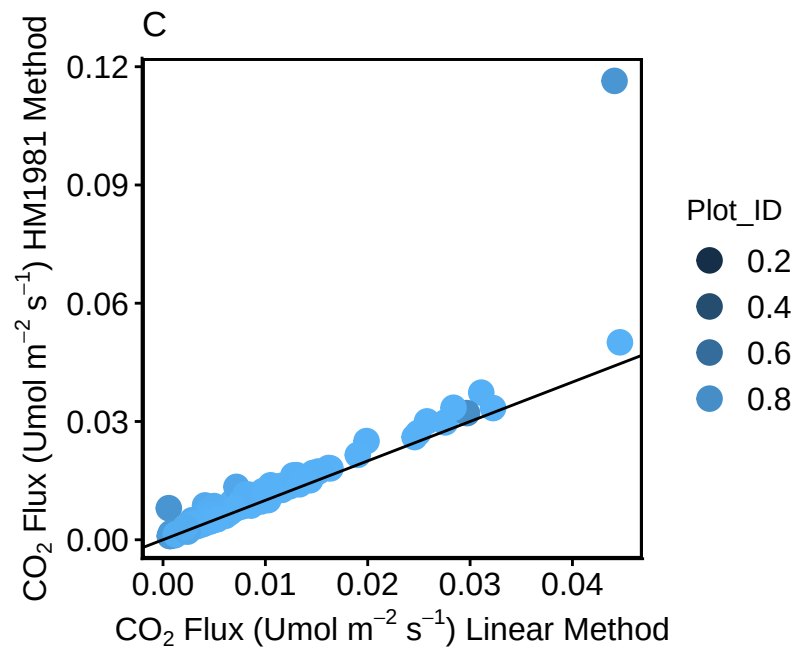
p3 <- ggplot(allflux_filtered, aes(lin_flux.estimate, HM81_flux.estimate, color = lin_r.squa
geom_point(size=4) + geom_abline() +
theme_classic() +
#xlim(0,50) + #ylim(0,50) +
labs(title = "C") +
ylab(expression(paste( CO [2], " Flux (Umol m"-2* " s"-1*) HM1981 Method")))) +
xlab(expression(paste( CO [2], " Flux (Umol m"-2* " s"-1*) Linear Method")))) +
guides(color = guide_legend(title = "Plot_ID")) +
theme(legend.position="right") +

```

```
theme(axis.title.x = element_text(size=12), axis.text = element_text(size=12),
      axis.title.y = element_text(size=12), legend.text=element_text(size=12),
      panel.border = element_rect(colour = "black", fill=NA, linewidth =1),
      aspect.ratio = 1)
```

p3

Warning: Removed 29 rows containing missing values or values outside the scale range (`geom\_point()`).



```
#ggarrange(p1, p2, p3, ncol=3, nrow=1, common.legend = TRUE, legend="none")
```

21 Read out final dataframe and export for use elsewhere

```
allflux_filtered_renamed <- fluxes_flagged %>%
  mutate(
    Date = mdy(str_extract(Plot, "[0-9]+/[0-9]+/[0-9]{4}")),
    Month = month.abb[month(Date)],
    Site = str_extract(Plot, "(PTR|OWC|CRC)"),
```

```

Zone = str_extract(Plot, "_TR_|_UP_") %>%
  str_remove_all("_"),
Replicate = as.integer(
  str_extract(Plot, "(?<=_ (TR|UP)_)[0-9]+")
),
Row_ID = as.integer(
  str_extract(Plot, "[0-9]+$")) ) %>%

# Optional: rebuild a clean Plot label
mutate(
  Plot_clean = paste(
    format(Date, "%Y-%m-%d"),
    Site,
    Zone,
    Replicate,
    Row_ID,
    sep = "_"
  )
) %>%

select(
  Plot_clean,
  Date,
  Month,
  Site,
  Zone,
  Replicate,
  Row_ID,
  everything(),
  -Plot
)

#export as a csv to the folder that you want
write.csv(allflux_filtered_renamed, file="Processed Data/COMPASS_Synoptic_LE_TreeStem_Fluxes.csv")
# #Change file name

```